

OceanEmbed: Solution Design and Innovation Roadmap

Reconstructing ocean temperature from the surface to 1000 m using satellite inputs. It covers what the repo does today, what is missing, and the features that would set it apart.

In one paragraph. OceanEmbed aims to infer ocean temperature at 15 depths, from the surface to 1000 m, using surface data from satellites. The data pipeline is real. It produces a 24 MB `dataset.nc` for the Bay of Bengal covering 60 days (Feb–Mar 2024), with 7 input channels and 15 target depths. The dashboard is polished, with six views: a Leaflet map, a depth profile, ARGO validation, ocean indices, a 3D landing page and a reconstruction page. But **no model has been trained yet, and every number in the UI is generated or hard-coded**. That includes the ARGO “ground truth” and its RMSE. The biggest gains come from three steps. First, train an honest model that is not fed inputs from the same simulation it predicts, and that reports its uncertainty. Second, validate it against real ARGO floats. Third, turn the reconstructed ocean into decisions: *cyclone intensification, marine heatwaves, fishing zones and sonar conditions*.

60 × 57 × 81

days × lat × lon on a
0.25° grid (8–22°N, 80–
100°E)

7 → 15

surface input channels
→ temperature at 15
depths

0

trained models in the
repo (HANDOFF.md: “not
started”)

49%

of the 1000 m target
values are fill, not ocean
(see §3)

1. Problem statement

Temperature below the surface drives cyclone intensification, the monsoon, fish distribution and underwater sound. Satellites only see the surface. The ocean interior is sampled sparsely, by roughly a few hundred ARGO floats in the North Indian Ocean and occasional research cruises. The challenge (SIH 26066) is to **learn a mapping from dense, daily satellite surface fields to full subsurface temperature profiles**. The result should fill the gaps between floats, every day, at 0.25° resolution.

- **INCOIS / IMD:** cyclone heat potential ahead of storms in the Bay of Bengal, which are frequent and intensify quickly.
- **Fisheries:** thermocline depth and fronts for Potential Fishing Zone (PFZ) advisories.
- **Navy:** sound-speed profiles, sonic layer depth and acoustic shadow zones.
- **Climate and ecosystems:** marine heatwaves below the surface, and coral stress around the Andaman and Lakshadweep islands.

2. Current architecture

The Python pipeline and the React front end are **not connected**. The dataset feeds nothing yet, and the dashboard builds its values from a seeded sine function in `oceanPhysics.ts`.

Today: the real data pipeline and the dashboard are not connected

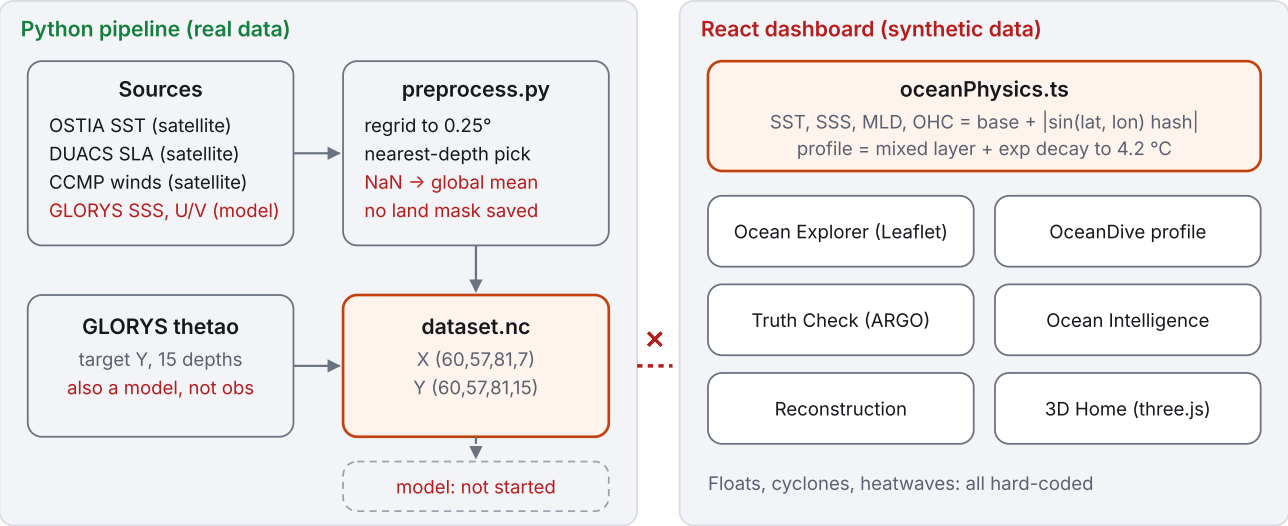


Figure 1. Current state. Red text marks the problems found in the code and data.

Technology stack

Layer	What is used
Front end	React 18 + TypeScript, Vite 6, Tailwind 3, Leaflet 1.9 (Stadia dark, Esri imagery, OSM), Recharts, three.js with @react-three/fiber and drei, lucide-react
Data pipeline	Python 3, xarray, NumPy, netCDF4, requests, the Copernicus Marine toolbox, and NASA Earthdata for CCMP (streamed and subset in memory)
Data	data/processed/dataset.nc (24 MB, float32). Raw NetCDFs are git-ignored.
Docs	A detailed README, and HANDOFF.md with the schema, provenance, caveats and next steps for training

3. What works, and what is missing

Working today

- A reproducible download and preprocessing pipeline with clear provenance for each variable, documented in HANDOFF.md.
- The target data is realistic. Surface (0 m) temperatures range from 20.4 to 33.2 °C, and 1000 m temperatures from 6.6 to 8.1 °C. SST correlates with the 0 m target at r = 0.90.
- A polished UI: basin switching with fly-to, a depth scrubber synced to the map point, an uncertainty band, float, cyclone and heatwave overlays, and a landing page with 3D bathymetry.
- The physics narrative is correct: the Bay of Bengal has a freshwater barrier layer and a shallow mixed layer, and the Arabian Sea has saltier surface water and a deeper one.

Gaps found in the code and data

Gap	Evidence	Impact
-----	----------	--------

No model	There is no <code>train.py</code> , no weights and no inference code. HANDOFF.md says "Model training not started". Deep learning is the core of the problem statement.	Blocker
The ARGO truth check checks nothing	In <code>getDepthProfile()</code> , <code>argoTemp = temp + sin(depth/35) * 0.22 + ...</code> . The "ground truth" is the fake prediction plus a sine wave, so the RMSE, bias and correlation on screen measure only that sine. Float IDs and battery levels are hard-coded.	Critical
All UI values are synthetic	SST, SSS, MLD, OHC and stratification come from <code>sin(lat * 12.9898 + lng * 78.233)</code> . The cyclones ("Deep Depression 03B", "ARB-01"), heatwaves and the Reconstruction page's input and output values are constants.	High
Fill values would corrupt training	<code>fill_nan</code> replaces land and seafloor NaNs with the global mean, and no mask is saved. At 1000 m, 135,360 of 277,020 values (49%) are the same fill value. At 0 m, 1,435 of 4,617 grid cells never change over time.	High
Target leakage	SSS and surface currents come from GLORYS12, the same model that provides the target temperature. The network can learn GLORYS's internal consistency, which inflates its skill and breaks the "satellite-only" claim.	High
The target is a model	GLORYS is a reanalysis. Training on it is fine, but the skill that matters has to be measured on independent ARGO profiles.	Medium
Arabian Sea is claimed but has no data	The README advertises both basins, but <code>dataset.nc</code> covers only 8–22°N, 80–100°E. Arabian Sea views are entirely hand-typed.	Medium
60 days, one season	Feb–Mar 2024 is before the monsoon and outside the cyclone season. A time split within those 60 days will overstate how well the model generalises.	Medium
Nearest-level depths	The 15 depths snap to the nearest GLORYS level, for example 0.49 m for 0 m. Interpolating vertically would be more exact.	Low

4. Proposed target architecture

Keep the pipeline and the UI. Put a trained model in between, and serve its outputs through an API, so every number on screen traces back to data.

Target: satellite inputs, a trained model with uncertainty, validated on ARGO

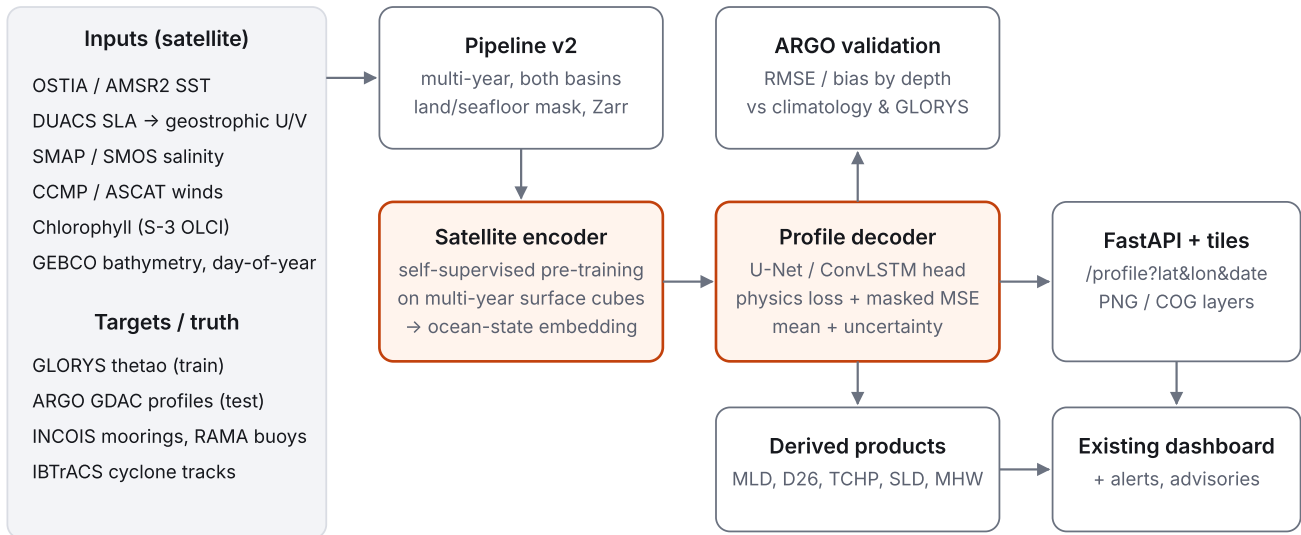


Figure 2. The “embedding” in the problem title becomes literal: a pre-trained encoder of satellite surface states feeds a profile decoder.

5. Features that would make it unique

Most teams will train a CNN on GLORYS and plot a map. The ideas below go further. They make the model trustworthy, honest about satellite-only inputs, and directly useful to INCOIS, IMD, fisheries and the Navy.

A. A model that can be defended

A1. True satellite embeddings, with self-supervised pre-training

Very high impact

Medium effort

- Pre-train an encoder, such as a masked autoencoder or a small vision transformer, on multi-year satellite surface cubes of SST, SLA, winds and salinity. This needs no subsurface labels at all.
- Fine-tune a light decoder that maps each embedding to 15-depth profiles. Compare it with a point-wise MLP, a plain U-Net and climatology, to show what the pre-training adds.
- A bonus feature is “find similar ocean days”: nearest-neighbour search in embedding space returns past states that look like today.

Why it stands out: it matches the problem title (“Satellite Embedding-Based”) literally, where most teams will only use the word.

A2. Physics-informed loss and profile constraints

High impact

Low-medium effort

- Use a masked MSE that ignores land and seafloor fill. Add penalties when temperature rises with depth below the mixed layer, and a smoothness term along depth.
- Optionally, predict coefficients of the main vertical patterns (EOFs) instead of 15 independent values. That gives smooth, physically shaped profiles with fewer parameters.

A3. Calibrated uncertainty on every pixel and depth

High impact

Low effort

- Use a deep ensemble or quantile outputs to produce a real $\pm\sigma$ that grows where the data is weak. The UI already draws a confidence band, so replace the hard-coded `0.25 + depth * 0.00085` with this.
- Show a reliability diagram against ARGO. When the model says 90% interval, the truth should fall inside it 90% of the time.

A4. Honest validation with real ARGO floats

Very high impact

Low-medium effort

- Download the actual profiles in the region from the Argo GDAC or ERDDAP, with quality flags. Match each one to the reconstruction for the same day and grid cell.
- Report RMSE and bias by depth against three baselines: WOA climatology, persistence, and GLORYS itself. Hold out whole floats, and a later season, so the test does not leak.
- Rebuild the Truth Check page on these results, with a selectable float and its real WMO ID and track.

Why it stands out: most competing dashboards will show fabricated metrics. A model that beats climatology on real floats is the strongest claim the team can make.

A5. Explainability: what drives each depth

Medium impact

Low effort

- Use integrated gradients or input ablation to show, for example, that SLA drives thermocline depth and SST drives the mixed layer. Show a depth-by-input importance heatmap in the Reconstruction view, replacing its hard-coded “embedding layer” graphics.

B. Applications built on the reconstruction

B1. Cyclone intensification watch (TCHP and D26)

Very high impact

Medium effort

- From each reconstructed profile, compute the depth of the 26 °C isotherm (D26) and the tropical cyclone heat potential (TCHP, in kJ/cm²). Map them daily.
- Overlay real IBTrACS or IMD tracks. For a demo, run a case study of a real Bay of Bengal cyclone (for example Remal, May 2024), showing TCHP along its path before landfall. The data window would need to be extended to cover it.
- Alert when a forecast track crosses water with TCHP above a threshold, a known signal of rapid intensification.

Why it stands out: it connects the model to INCOIS's and IMD's operational question, and replaces the hard-coded "Deep Depression 03B".

B2. Subsurface marine heatwave detector

High impact

Medium effort

- Apply the standard Hobday definition: above the 90th-percentile climatology for at least 5 days. Apply it *at depth*, not only at the surface. Heatwaves hidden beneath a normal-looking surface are a current research topic.
- Add a coral-stress layer for the Andaman and Lakshadweep islands based on degree heating weeks at 10–30 m.

B3. Thermocline-aware Potential Fishing Zones

High impact

Medium effort

- INCOIS PFZ advisories rely on surface SST fronts and chlorophyll. Adding reconstructed thermocline depth helps predict where deeper-feeding species such as tuna will be.
- Deliver advisories as SMS or app messages in Tamil, Telugu, Odia, Bengali and Malayalam, as coastal sectors with distance and bearing from a landing centre.

B4. Acoustic ocean: sound-speed profiles and sonar range

High impact

Low-medium effort

- Compute sound speed from temperature, salinity and depth with a standard empirical formula such as Mackenzie's. Derive the sonic layer depth, the shadow zones and a simple detection-range map.

Why it stands out: this is a naval and strategic use that almost no SIH team will think of, and it costs little once profiles exist.

C. Operations and experience

C1. Daily near-real-time run and 1–7 day forecast

High impact

Medium effort

- Run a scheduled job that pulls yesterday's satellite fields and publishes the reconstruction. Next, add a forecast mode that takes a sequence of surface fields and predicts the subsurface state days ahead.
- When new ARGO profiles arrive, blend them in near the float with a simple optimal-interpolation correction, and show the effect.

C2. Data-backed dashboard and a 3D ocean volume

Medium impact

Low–medium effort

- Swap `oceanPhysics.ts` for API calls. Keep the same types, so the components barely change.
- Use the `three.js` setup that already exists to render the reconstructed cube as a volume. Add depth slices and an isotherm surface such as 26 °C that the user can rotate.

C3. Ocean copilot and auto bulletins

High demo impact

Low effort

- Answer questions such as “where is the thermocline shallower than 40 m in the northern Bay this week?” and apply the answer to the map. Generate a daily PDF bulletin with TCHP, heatwaves and PFZ zones.

6. Prioritisation

Do the fixes in Phase 0 first. Without them, any metric the team shows can be challenged. Then take the top-left quadrant.

A real ARGO benchmark and uncertainty give the most credibility per hour

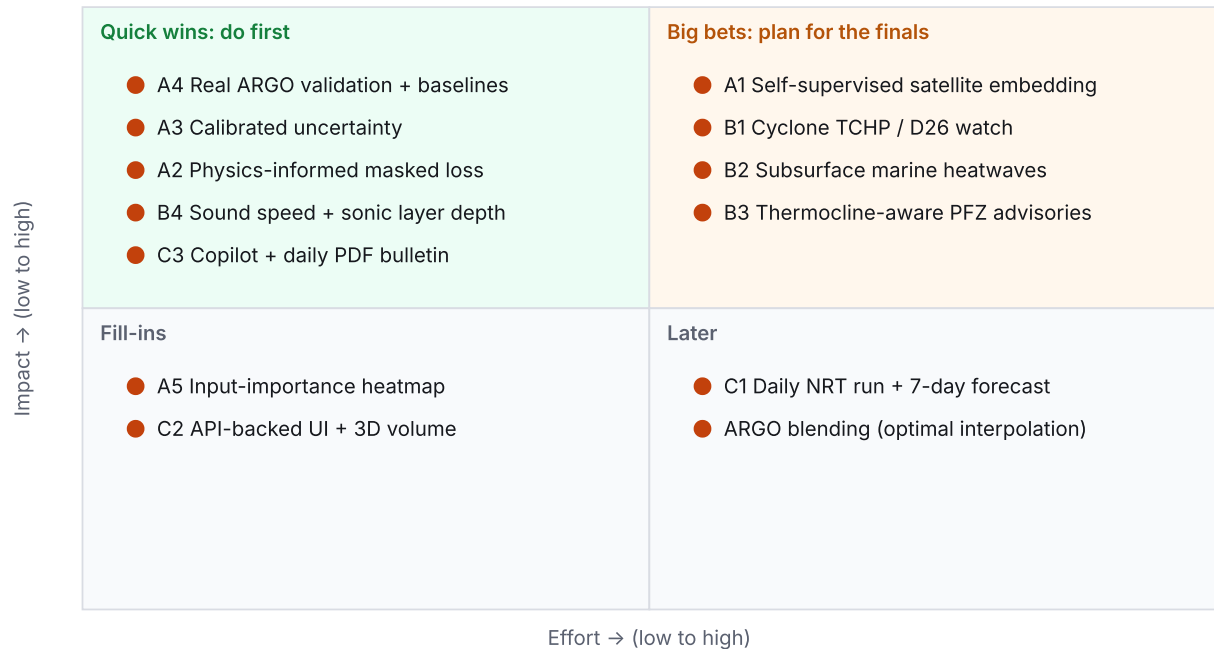


Figure 3. Impact vs effort. Positions within a quadrant are not to scale.

Suggested build order

Phase	Work	Outcome
0 · Fix the data ~2 days	Save land and seafloor masks in <code>dataset.nc</code> instead of mean-filling. Replace GLORYS SSS and currents with SMAP/SMOS salinity and geostrophic currents from DUACS. Extend to both basins and at least 3 years so every season is covered. Interpolate depths.	Training data that is clean, satellite-only and matches what the README claims.
1 · Baselines + truth ~1 week	WOA climatology and MLP baselines, U-Net with masked physics loss (A2), ARGO matchups (A4), ensemble uncertainty (A3).	Real RMSE-by-depth numbers that beat climatology.
2 · Wire the UI ~1 week	FastAPI serving profiles and tiles. Replace <code>oceanPhysics.ts</code> and rebuild the Truth Check page on real floats. Add D26/TCHP (B1) and sound speed (B4).	Every number on screen traces back to data or the model.
3 · Differentiate ~1–2 weeks	Self-supervised embedding (A1), subsurface heatwaves (B2), a cyclone case study, PFZ advisories (B3), bulletins (C3).	A story with genuine research value and a clear operational use.

4 · Operate stretch	Daily NRT run, forecast mode and ARGO blending (C1), 3D volume (C2), regional-language alerts.	A prototype of an operational INCOIS service.
-------------------------------	--	---

7. The one-line pitch

OceanEmbed turns what satellites see at the surface into a daily 3D picture of the Indian Ocean down to 1000 m, with honest error bars and proof against ARGO floats. From that picture it derives cyclone fuel, hidden marine heatwaves, fishing zones and sonar conditions.

Success metrics to show judges

- RMSE and bias by depth against held-out ARGO profiles, compared with WOA climatology and GLORYS. Headline number: the % improvement over climatology in the 50–200 m thermocline.
- Calibration: the share of ARGO observations that fall inside the model's 90% interval (target: about 90%).
- Cyclone case study: TCHP along the track against the observed intensity change.
- Latency from satellite data availability to a published reconstruction.

Dataset statistics in §3 were computed directly from `data/processed/dataset.nc`. External products, thresholds and cyclone references are from general knowledge. Verify them against Copernicus Marine, Argo, IBTrACS and INCOIS documentation before relying on them.